Preparation of a Software Project Plan Covering Scope, Timeline and Team Roles for a Sample Application

Case Study 2

Aim

To prepare a software project plan that defines the scope, timeline, team roles, risks and deliverables for a sample application.

Objective

- Define the project scope, including inclusions and exclusions.
- Break the work into phases with a realistic timeline.
- Assign roles and responsibilities to team members.
- Identify project risks and plan mitigation.
- Define deliverables, milestones and success criteria.

Selected Sample Application

Hospital Appointment Booking System – a web and mobile application that lets patients book, reschedule and cancel doctor appointments online, and lets hospital staff manage doctor schedules.

1. Project Scope

In Scope

- Patient registration and login.
- Search doctors by department, name or availability.
- Book, reschedule and cancel appointments.
- Email / SMS appointment reminders.
- Doctor schedule management and admin dashboard.
- Online payment of consultation fees.

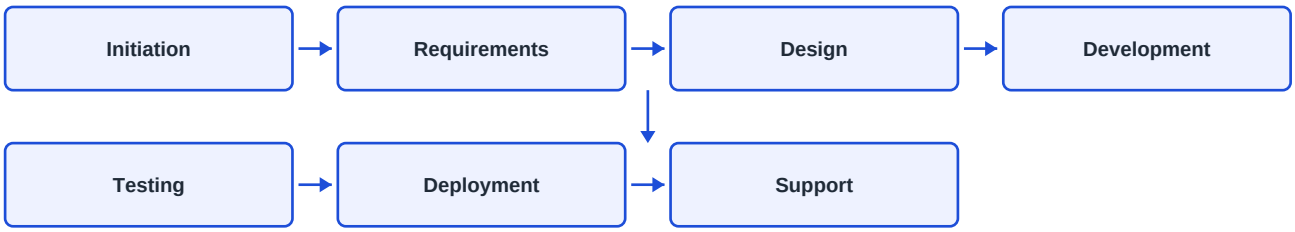
Out of Scope

- Electronic Medical Records (EMR) and prescriptions.
- Pharmacy and laboratory integration.
- Video consultation (planned for a future release).

Assumptions and Constraints

- Payment will use a third-party gateway.
- Fixed team of 6 members; total duration of 12 weeks.
- Application hosted on a cloud platform.

2. Work Breakdown Structure



3. Project Timeline (12 Weeks)

Phase	Activities	Duration	Milestone
Initiation & Planning	Feasibility, scope, project plan	Week 1	Plan approved
Requirement Analysis	SRS, user stories, prioritization	Weeks 2–3	SRS signed off
System Design	Architecture, UI wireframes, database schema	Weeks 3–4	Design approved
Development – Sprint 1	Registration, login, doctor search	Weeks 5–6	Demo 1
Development – Sprint 2	Booking, rescheduling, cancellation	Weeks 7–8	Demo 2
Development – Sprint 3	Payment, notifications, admin dashboard	Weeks 9–10	Feature complete
Testing & UAT	System testing, UAT, bug fixing	Week 11	Test sign-off
Deployment & Handover	Go-live, training, documentation	Week 12	Go-live

4. Team Roles and Responsibilities

Role	Count	Responsibilities
Project Manager	1	Planning, scheduling, risk management, stakeholder communication
Business Analyst	1	Gather requirements, write SRS and user stories
UI/UX Designer	1	Wireframes, prototypes, usability
Developers (Full-stack)	2	Front-end, back-end, API and database development
QA / Test Engineer	1	Test cases, execution, defect reporting

5. Deliverables

- Project Plan and SRS document.
- Design documents (wireframes, ER diagram, architecture).
- Working application (web and mobile) with source code.
- Test reports and user manual.

6. Risk Management

Risk	Impact	Mitigation
Requirement changes	High	Agile sprints, change control process
Team member unavailability	Medium	Cross-training, documented code
Payment gateway delay	Medium	Early integration with sandbox account
Security / data privacy issues	High	Encryption, role-based access, security testing

7. Communication and Tracking

Daily 15-minute stand-ups, sprint reviews and retrospectives every two weeks, and weekly status reports. Progress is tracked on a task board (e.g., Jira / Trello) using burndown charts.

Results

- A complete, time-boxed 12-week plan with clear milestones was prepared.
- Every team member has well-defined responsibilities, avoiding overlap.
- Scope boundaries and risks were identified early, reducing the chance of scope creep.

Conclusion

A well-prepared software project plan provides a roadmap for the team, aligns stakeholder expectations and improves the chances of delivering on time and within budget. The Hospital Appointment Booking System plan shows how scope, timeline and roles work together in practice.